

Transform your Smartphone in a Real-time Sonagram Player

Jean-Denis DUROU
The University of Toulouse
Toulouse, France
durou@irit.fr

Gilles AZZARO
Independent
Toulouse, France
azzarogilles@gmail.com

Achille JEANVOINE
The University of Toulouse
Toulouse, France
achille.jeanvoine@etu.toulouse-inp.fr

Jean MÉLOU
The University of Toulouse
Toulouse, France
jean.melou@toulouse-inp.fr

Hugo PAUGET BALLESTEROS
Sorbonne Université
Paris, France
hugopauget@gmail.com

Clément LACIRE
The University of Toulouse
Toulouse, France
clement.lacire@etu.toulouse-inp.fr

Julie REMENANT
The University of Toulouse
Toulouse, France
julie.remenant@etu.toulouse-inp.fr

Yvain QUÉAU
The University of Caen
Caen, France
yvain.queau@ensicaen.fr

Gabriel GOURNAY
The University of Toulouse
Toulouse, France
gabriel.gournay@etu.toulouse-inp.fr

Floriane PAYEN
The University of Toulouse
Toulouse, France
floriane.payen@etu.toulouse-inp.fr

Abstract

In this article, we present a multimedia system for reading a sonagram using a smartphone, the two main features of which are that it is both interactive and real-time.

Keywords

Smartphone, Sonagram, Audio Player, Interactive, Real-time.

ACM Reference Format:

Jean-Denis DUROU, Jean MÉLOU, Yvain QUÉAU, Gilles AZZARO, Hugo PAUGET BALLESTEROS, Gabriel GOURNAY, Achille JEANVOINE, Clément LACIRE, Floriane PAYEN, and Julie REMENANT. 2025. Transform your Smartphone in a Real-time Sonagram Player. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3756141>

Recording a Sound and Playing it Back

Recording sounds: this dream was first realised by a Frenchman named Édouard-Léon Scott de Martinville. His device, called the *phonograph*, enabled the first material transcription of the voice around 1860. However, it was only in 2007 that four American researchers at Berkeley succeeded in replaying several of Scott de Martinville's *phonographs* [3], 150 years after they were recorded, because recording a sound and playing it back present very different levels of difficulty.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '25, Dublin, Ireland

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3756141>

Playback of a sound recording was only possible in 1877, with two simultaneous inventions: Thomas Edison's *phonograph* and Charles Cros' *paléophone*. So it was only almost twenty years later than the first recording that sound could be, for the first time, recorded and played back. Since then, recording and playback equipment has continued to evolve, and is now increasingly processed entirely digitally, from recording to playback.

Representing Sounds as Sonagrams

The oldest time-frequency representation of a sound is the musical score, which is not enough to play back a piece of music *automatically*, since it has to be *interpreted*. A sonagram is a more faithful time-frequency representation of a sound. However, it too does not contain the entire signal. Indeed, if the Fourier transform of the signal is calculated for each position of a sliding window, only the complex modulus of this transform is retranscribed. In other words, the phase is lost.

French artist Gilles Azzaro [1] describes himself as a “voice sculptor”. A sonagram can be represented in 3D by adding the complex modulus to its two original dimensions (time and frequency). This 3D representation is thus particularly suited to 3D printing. Thanks to some modifications, Gilles Azzaro knows how to transform it into a real artwork (see Figure 1). Is it possible to restore the encoded sound in such a representation? That is the challenge we set ourselves, together with Gilles Azzaro.

Phase Retrieval

In a previous edition of the ACM Multimedia Art Exhibition [5], we already succeeded in achieving this objective. From the outset, we restricted ourselves to 2D sonagrams, rather than 3D, to simplify the process of capturing the images using a smartphone. Once captured, the main obstacle to reading the sonagram was that the

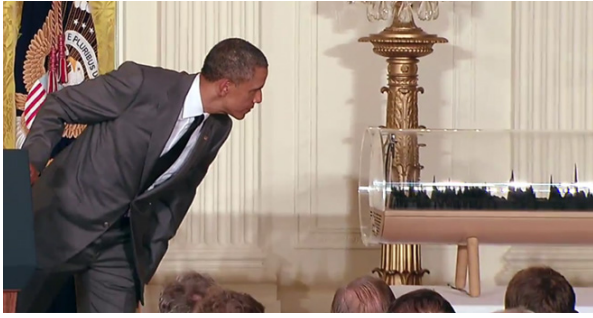


Figure 1: Sonagram of a speech by former US President Barack Obama, 3D printed by French artist Gilles Azzaro [1], exhibited in 2014 at the White House.

phase is lost, as already said. This results in distortions in the reconstructed sound. A video by Gilles Azzaro (<https://www.youtube.com/watch?v=kXANN0Nfpeo>), in which a sonagram produced using the PhonoPaper application [6] is played back, highlights such distortions.

To overcome this obstacle and make the reconstructed sound audible, we used in [5] the phase reconstruction method proposed in 1984 by Griffin and Lim [4], originally designed to reconstruct the phase in cases where the sonagram of the original signal has been modified, for example to improve sound quality. In this case, the original phase is no longer consistent with the sonagram. Although the results were satisfactory, given that this method is iterative, the interactive and real-time aspect could not really be achieved.

In this new opus, we have opted for another method of phase reconstruction: the SPSI algorithm (*single pass spectrogram inversion*), proposed by Beauregard et al. [2]. It is a rather fast non-iterative algorithm, but its greatest advantage is that, to calculate the phase values at time t , it uses only the phase values at time $t - 1$. This algorithm is therefore much better suited to our application than the one we used in [5], as it allows us to process the sonagram in small parts, and therefore to carry out processing in real-time. Only drawback: less well-known than that by Griffin and Lim [4], the SPSI algorithm was not implemented in the libraries we use.

Technical Features

The decision to develop a mobile application has reduced the choice of available programming languages, libraries and compilation systems, many of which do not offer a mobile version. Furthermore, the need for efficiency and speed is not to the advantage of cell phones, which are generally less powerful than computers.

On the other hand, the idea of creating a web application via a server, i.e. with remote processing, was mooted, but quickly abandoned, as the delay in sending/receiving information between the phone and the server risked being too long and compromising the real-time aspect of the application.

We therefore chose to develop the application on Android Studio (<https://developer.android.com/studio>), which is the official IDE for developing mobile applications on the Android operating system. The drawback of this choice is that the proposed application only

works on Android-equipped phones, which excludes users of iOS, the exclusive operating system for Apple smartphones.

Proposed Multimedia System

A number of modifications of the sonagrams are needed by the application, in order to concentrate the relevant information:

- Only the positive frequencies are kept.
- The ordinate corresponds to the logarithm of the frequency, in order to spread out the low frequencies.
- The grey levels correspond to the square of the complex modulus, expressed in dB.

An example of such a sonagram is shown in Figure 2.

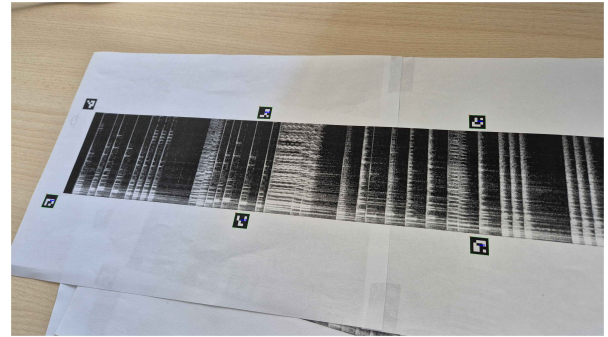


Figure 2: Example of sonagram which can be read by our interactive and real-time audio player (see attached video). Perspective causes a homographic distortion, which can be corrected thanks to regularly distributed ArUco markers.

To better evaluate the relevance of the proposed multimedia system, a video accompanies this submission. Any sound signal could be treated in the same way. In order to make the workshop fun, some visitors will be asked to briefly introduce themselves. Therefore, a collection of sonagrams can be displayed on a screen, and then read by other visitors belonging a smartphone.

An important peculiarity of the proposed multimedia system is to make it interactive and playful. In the version which will be presented at the exhibit, the user will have the control on the playback, giving the pace of lecture and also deciding on its direction (forward or backward), by simply moving his smartphone.

Conclusion and Perspectives

The main thrust of our previous attempt [5] was to reconstitute the phase “on the fly”, so as to be able, for instance, to read a sonagram printed on a sheet of paper, then crumple up the sheet and note the effect on the restored audio signal, as done in a video presentation of PhonoPaper (<https://www.youtube.com/watch?v=lzoVnqLy29U&t=0s>). We now know how to carry out the first step (the second stage can easily be tested on site).

Another type of 3D sonagrams exist, in the form of *lithophanes*: the 3D printing faces the light placed at the back and is seen by transparency. It turns out that the image produced is the same as if it had been printed on a plane. A perspective of our work is thus to print some famous sentences in 3D, and to constitute an exhibition of such lithophanes, beside the recreational workshop.

References

- [1] Gilles Azzaro. 2022. Homepage. <http://www.gillesazzaro.com>.
- [2] Gerald Beauregard, Mithila Harish, and Lonce Wyse. 2015. Single pass spectrogram inversion. In *Proceedings of the IEEE International Conference on Digital Signal Processing*. 427–431.
- [3] Patrick Feaster. 2017. Playback Methods for Phonogram Images on Paper. In *Proceedings of the 2016 Joint Technical Symposium*.
- [4] Daniel Griffin and Jae Lim. 1984. Signal estimation from modified short-time Fourier transform. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 32, 2 (1984), 236–243.
- [5] Hugo Pauget Ballesteros, Gilles Azzaro, Jean Mélou, Yvain Quéau, and Jean-Denis Durou. 2022. Read Your Voice – A Playful Interactive Sound Player. In *Proceedings of the 28th ACM International Conference on Multimedia*.
- [6] Alexander Zolotov. Latest release - 1.6 (2018-03-10). PhonoPaper. <http://www.warmplace.ru/soft/phonopaper/>.